Tooth Eruption

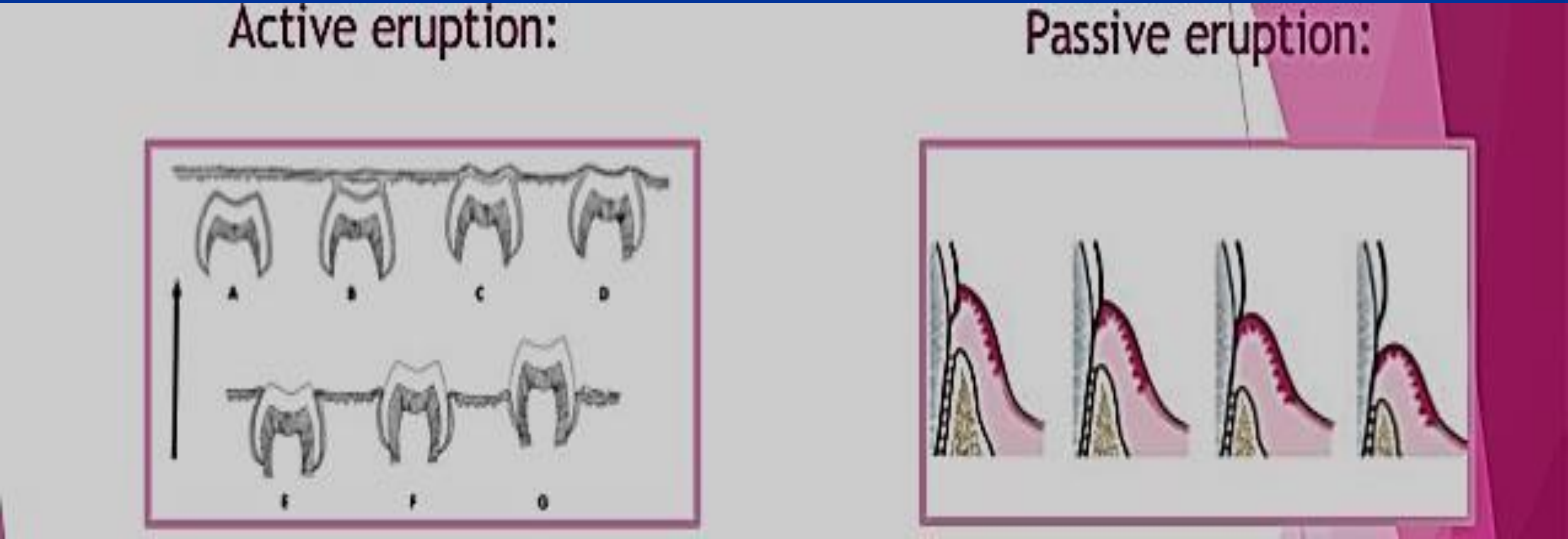
Learning outcomes of the lecture

At the end of this lecture the student should be able to:

- 1- Describe the three phases of tooth eruption: pre-eruptive , eruptive(pre-functional) & post-eruptive(functional)
- 2- Describe the accompanying changes that occur in the surrounding ,overlying &underlying tissues during eruption of the tooth.

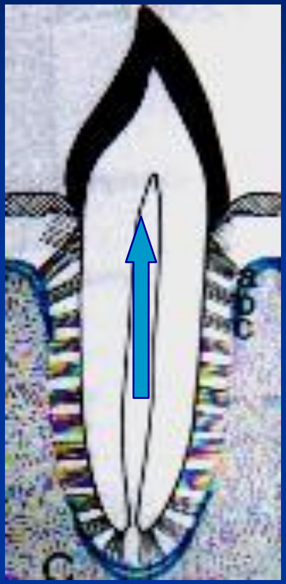
TERMINOLOGY OF ERUPTION

- **ACTIVE ERUPTION**: It is the actual movement of the tooth from its developmental site to its position in the dental arch.
- **PASSIVE ERUPTION**: Does not involve tooth movement but occurs due to apical recession of gingival tissue exposing more tooth structure into the oral cavity.



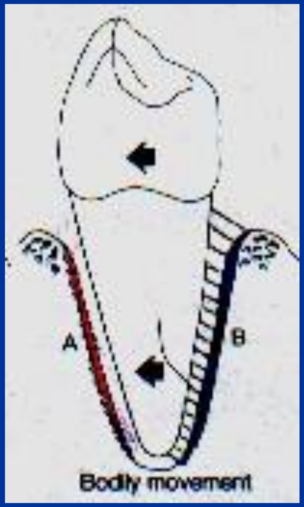
Types of Tooth Movement

Axial movement



Occlusal movement in the direction of the long axis of the tooth

Bodily movement

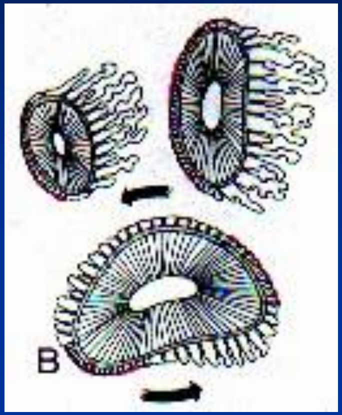


Movement to one direction mesial,distal Buccal or lingual

Tilting (tipping)

Movement around a transverse axis

Rotatory movement



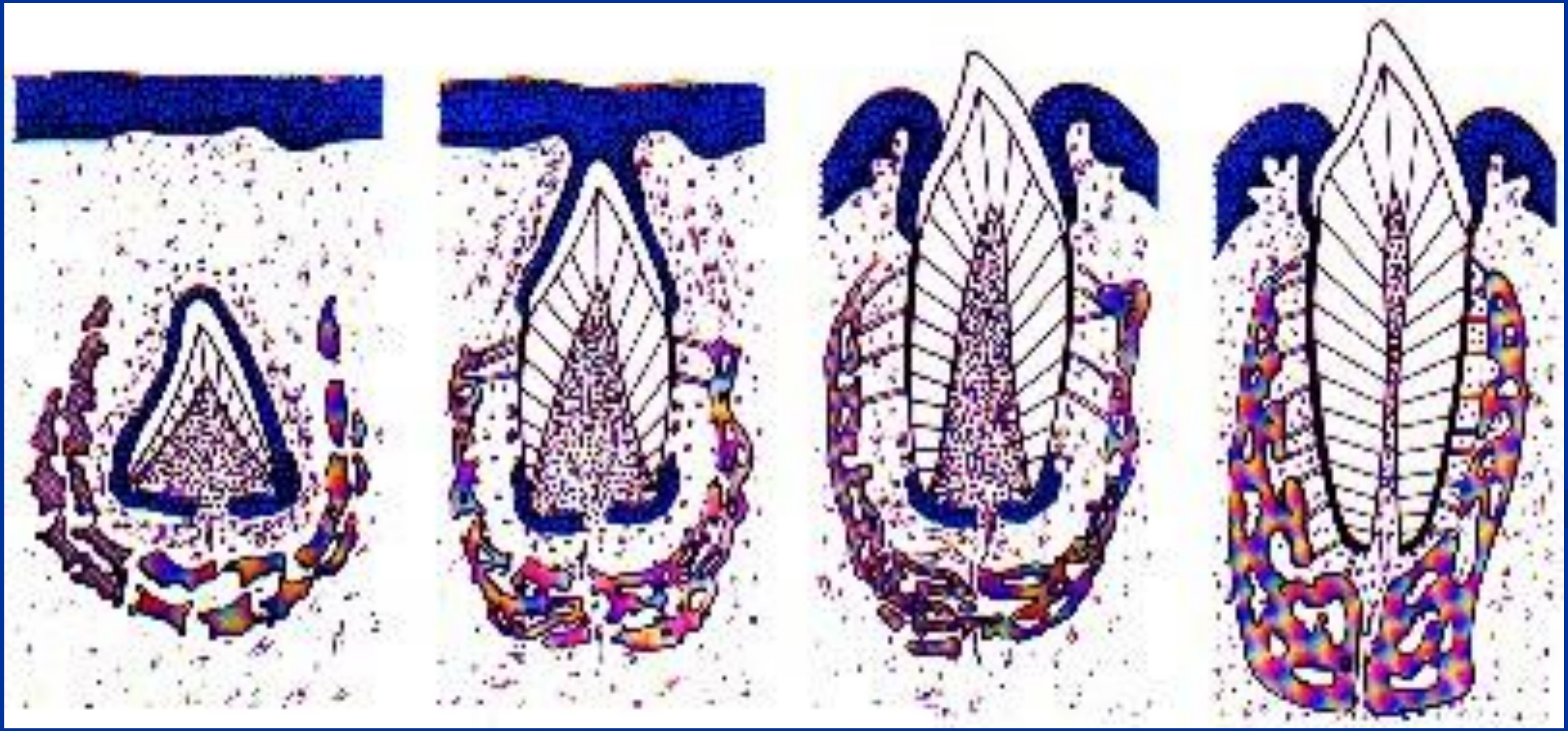
Movement around a longitudinal axis

Eruption of Teeth

Def: Axial or occlusal movement of teeth (or teeth germs)

From ⇨ Their developmental position in the jaw

To ⇨ Their functional position in the oral cavity



Phases of eruption:

I. Pre eruptive phase

II. Eruptive phase

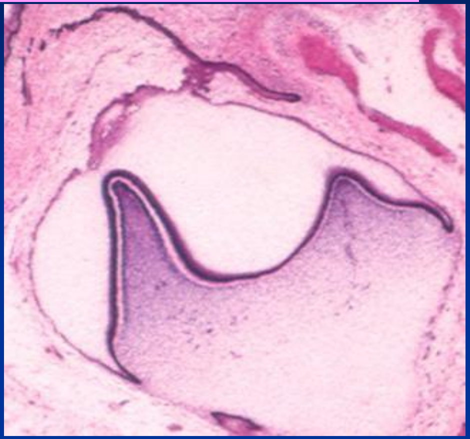
III. Post eruptive phase

Phases of Erption

In Any phase we should speak about:

- Phase
  - When it starts & ends
  - Ongoing events
  - Type of movement

1- Pre-eruptive phase:



- Begins…….from early phases of crown formation **(early bell stage)**
- Ends……….at the time of crown completion. **(At the beginning of root formation)**

- So, All the pre-eruptive movements occur within the crypts of developing crowns before root formation begins.

Pattern of movements in pre-eruptive phase

A- Pattern of tooth movement in the pre-eruptive phase for deciduous teeth

B- Pattern of tooth movement in the pre-eruptive phase for permanent teeth with deciduous predecessors (anteriors & premolars)

C - Pattern of tooth movement in the pre-eruptive phase for permanent teeth without deciduous predecessors(molars)

A- Pattern of tooth movement in the pre-eruptive phase for deciduous teeth:

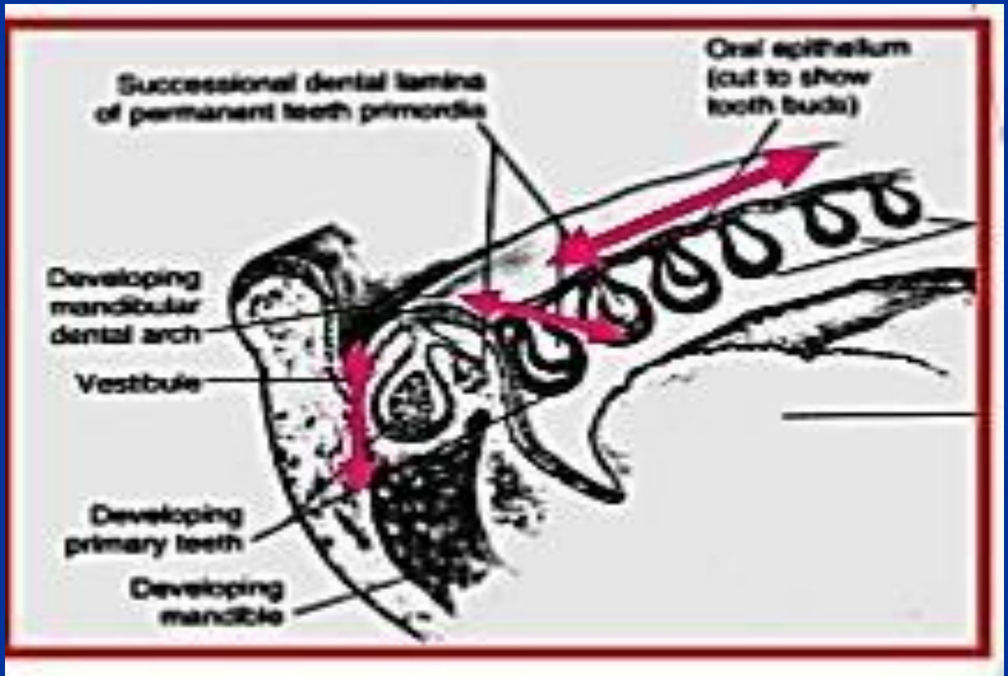
- At the beginning…….. deciduous tooth germs are small with enough space between them
- Then, Rapid growth of deciduous tooth germs utilized the available space…….. crowding occurs

This crowding is relieved by:

*Growth of jaw in length….(A,B,C)…..move forward(mesially)
(E)………..move backward (distally)
(Bodily movement)

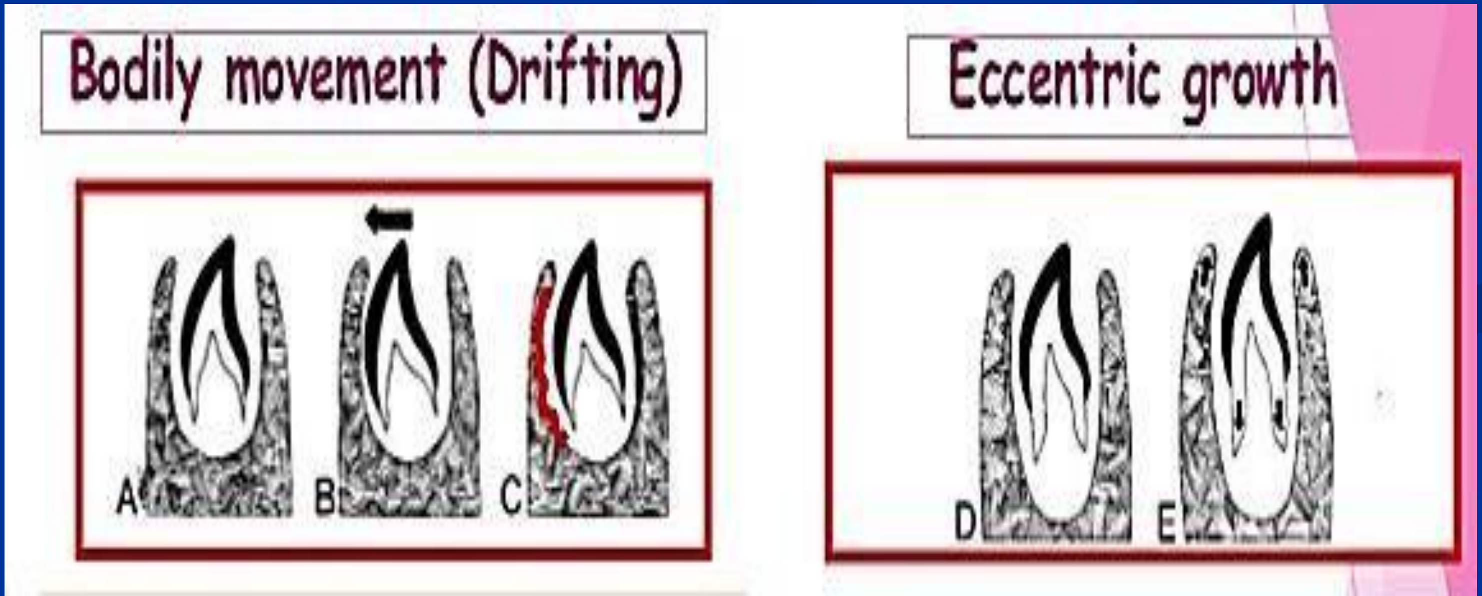
*Growth of jaw in width…all tooth germs….moved outward
(towards vestibule)
(Bodily movement)

*Growth of jaw in height…all tooth germs..moved occlusally
by (excentric growth)



Movements in the pre-eruptive stage:

- **1-Bodily movement:** There will be bone resorption in the surface toward which tooth moves. While bone deposition occurs on the crypt wall behind it.
- **2-Eccentric growth:** It means that one part of the developing tooth germ remains stationary and the remainder continues to grow leading to shift in its center.

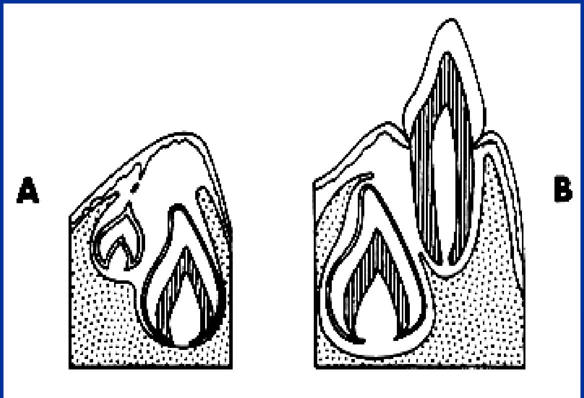


B-Pattern of tooth movement in the pre-eruptive phase for permanent teeth

with deciduous predecessors (anteriors & premolars)

1-Anteriors: (incisors and canines)

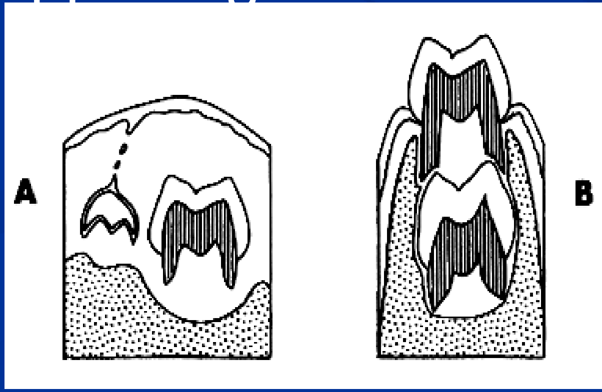
Their tooth germs begin development lingual to the incisal level of deciduous teeth, later after the deciduous erupts, they become positioned lingual to the the apical third of their roots



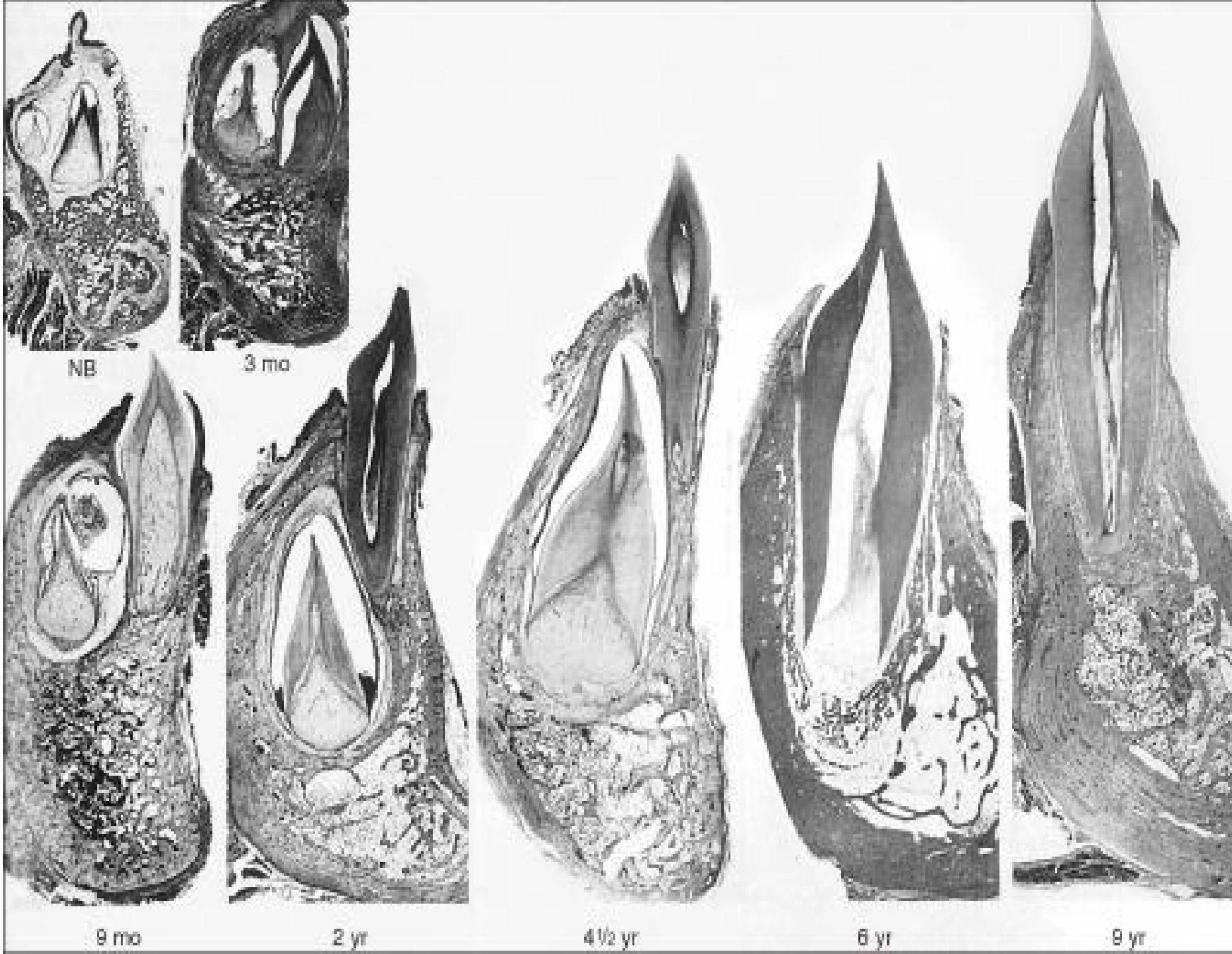
2- The premolars

-They shift from a position near the occlusal area of deciduous to a position enclosed within the roots of deciduous molars.

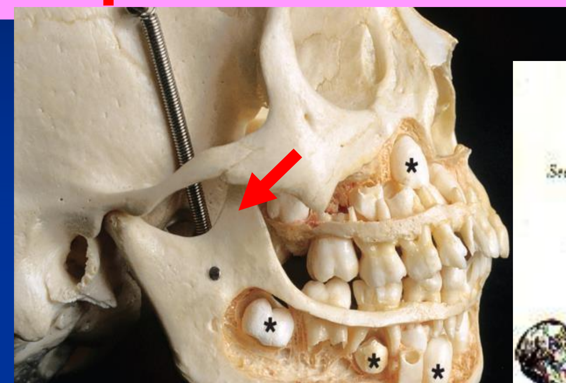
- This occur as a result of deciduous eruption & increase in the height of supporting structures

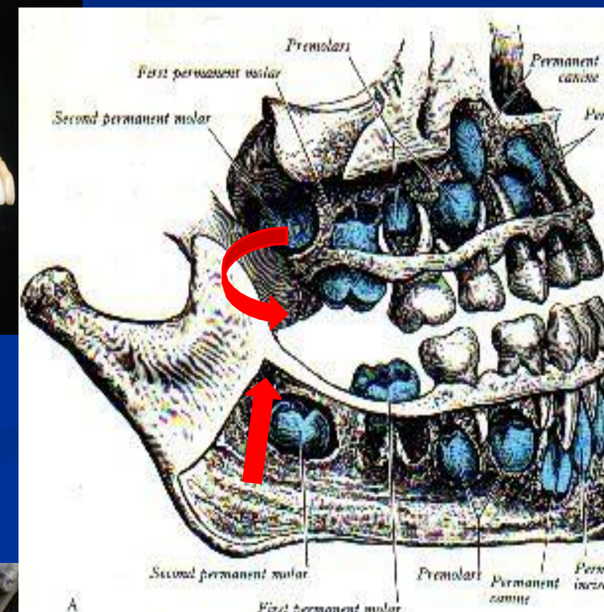


A



C- Pattern of tooth movement in the pre-eruptive phase for permanent teeth without deciduous predecessors(molars)

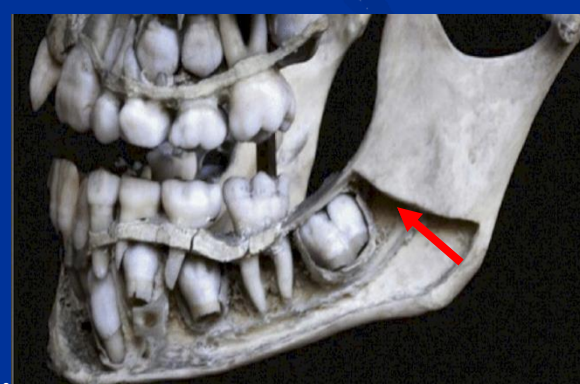




Permanent molars:

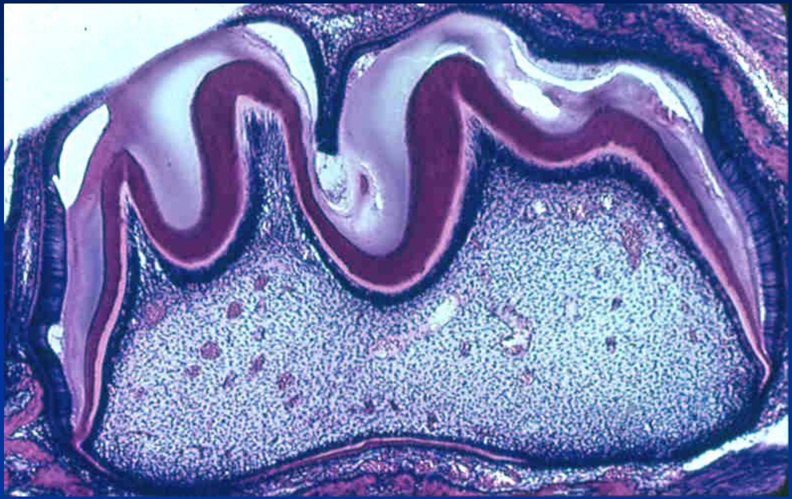
*Maxillary permanent molars develop in the maxillary tuberosities….. occlusal surfaces facing distally → then swing with growth of the maxilla

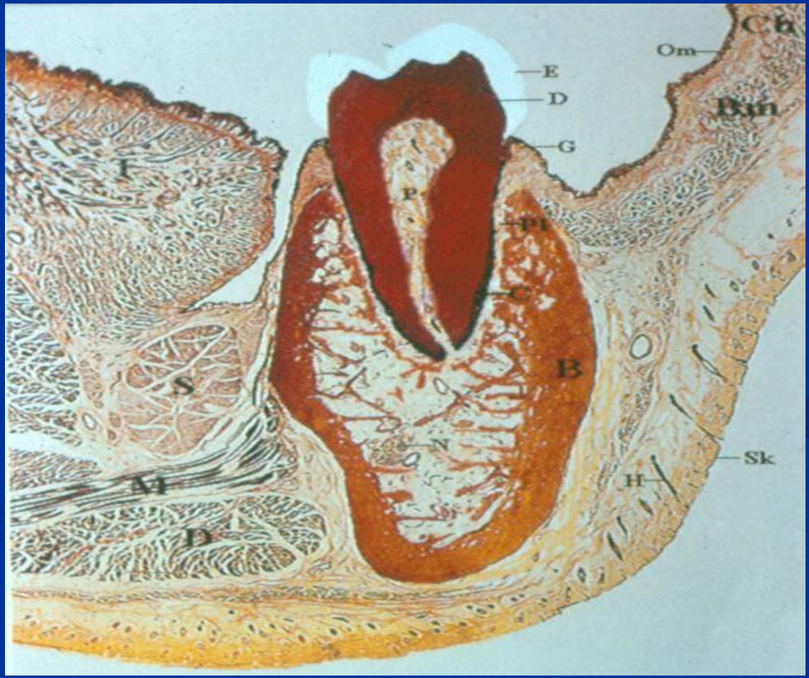
*Mandibular permanent molars develop in the ramus of the mandible with occlusal surfaces inclined mesially → then tilt to be upright with growth of the mandible



2) Eruptive (Prefunctional) phase:

- It starts at the beginning of root formation
- It ends when the tooth reaches occlusion
- <u>**N.B: the rate of tooth eruption is more than the rate of jaw growth**</u>





2) Eruptive (Prefunctional) phase:

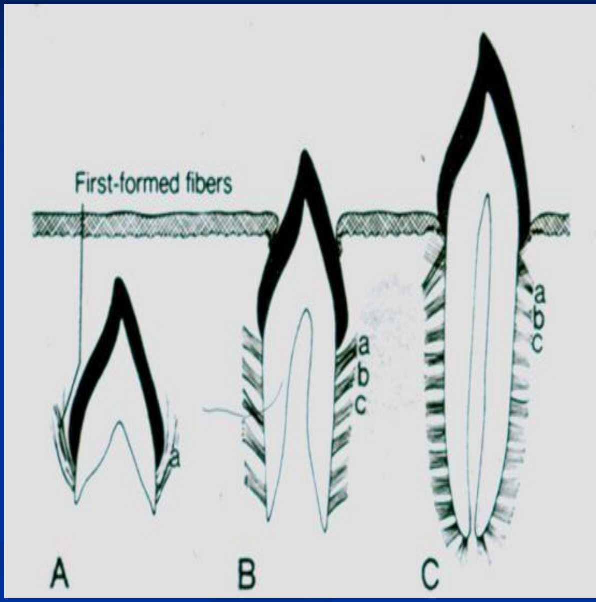
- It is characterized by <u>4 major events</u>

A) Formation of the root

B) Movement occur occlusally through the bony crypt to reach the oral mucosa.

C) Penetration of the crown tip through the epithelial layers allowing the enamel to enter oral cavity.

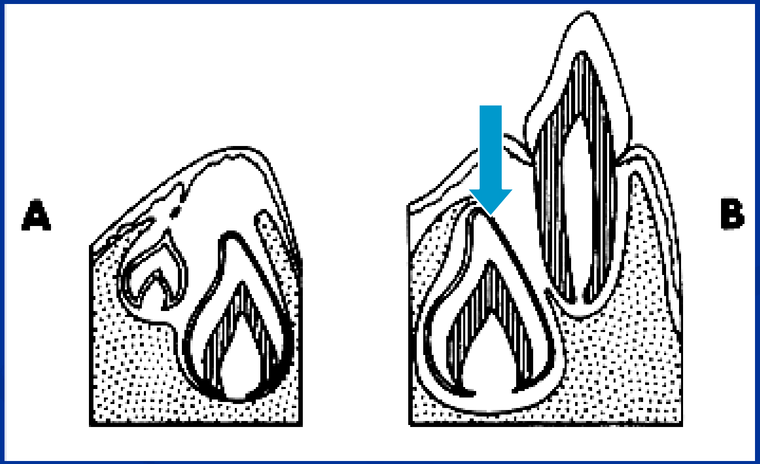
D) Intraoral occlusal movement of erupting tooth till contact opposing teeth



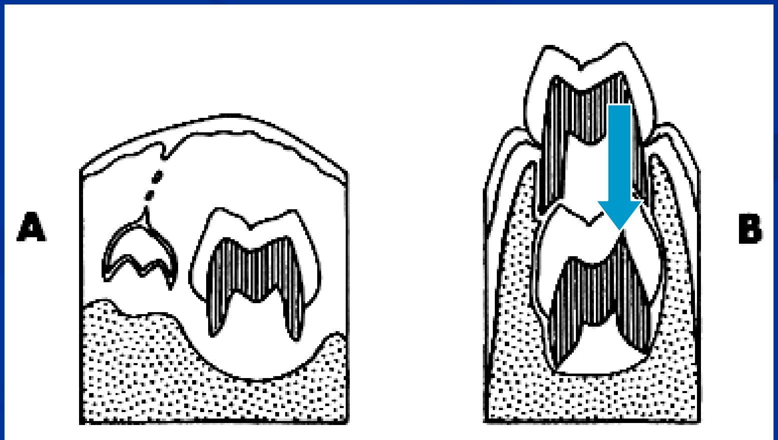
ERUPTION PATHWAY

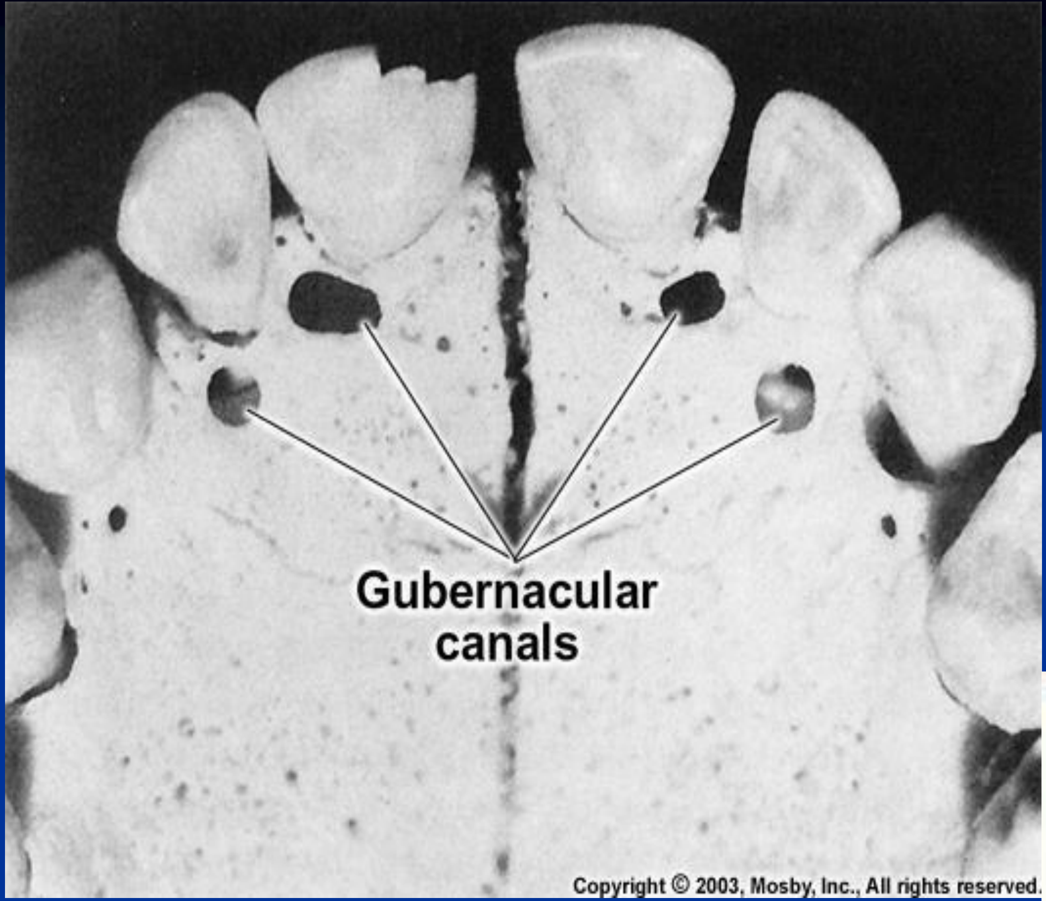
As the deciduous erupt the successor tooth germ lie in its own bony crypt which completely surrounded by bone except for a small canal known as <u>GUBERNACULAR CANAL</u> .

- In the anterior teeth, these canals are seen in the lingual plate of bone



- while in premolars the canals are present in the socket of the corresponding deciduous molars.





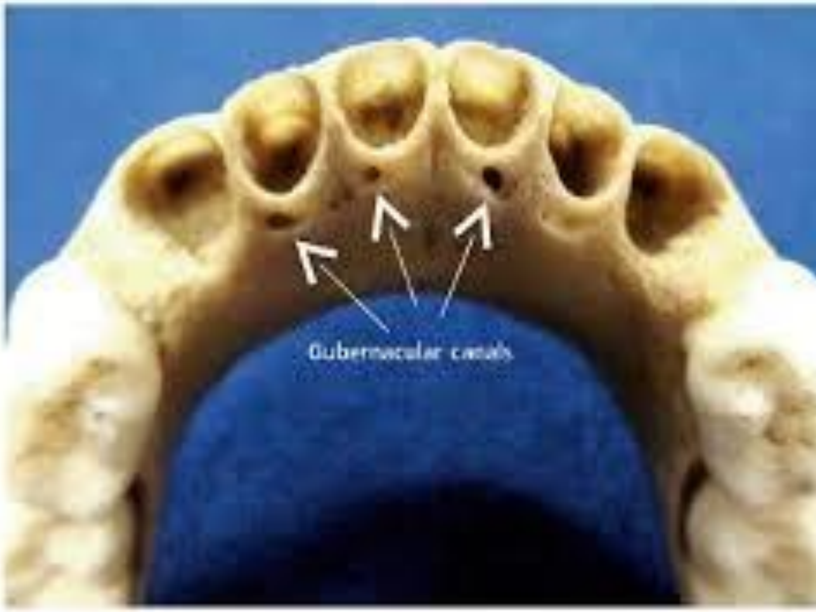
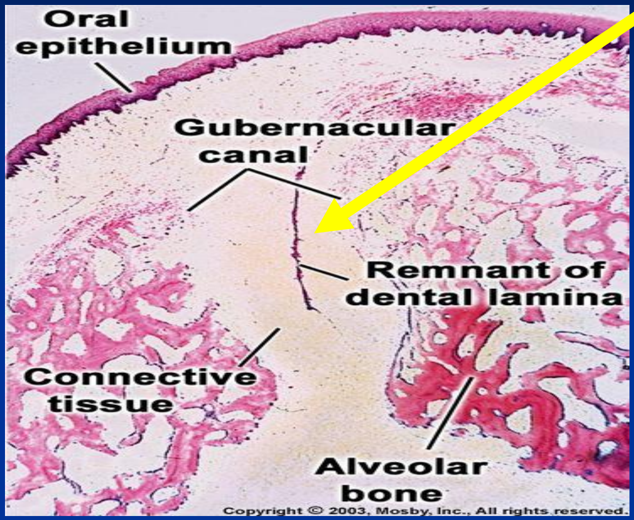
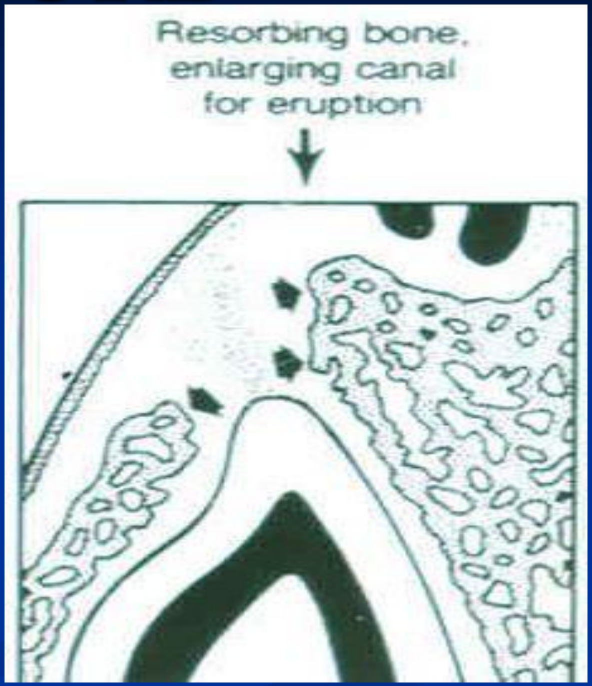


Figure 1 - Dry child skull. Gubernacular canals located in the alveolar bone crest, behind the mandibular deciduous incisors

GUBERNACULAR CORD



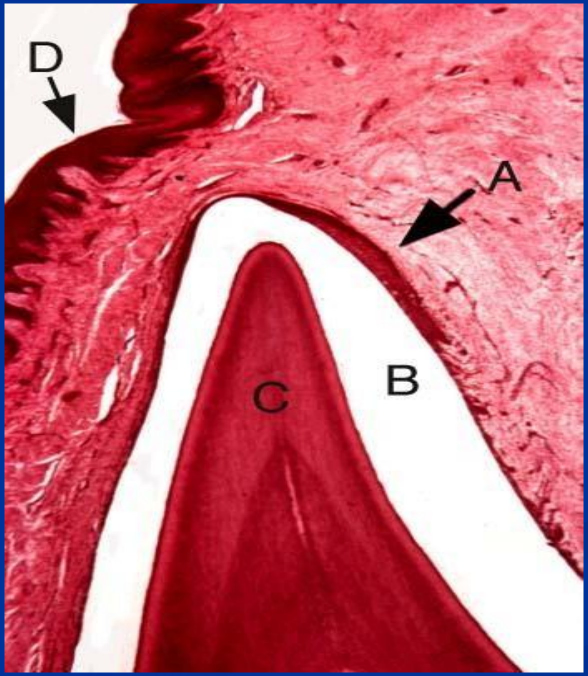


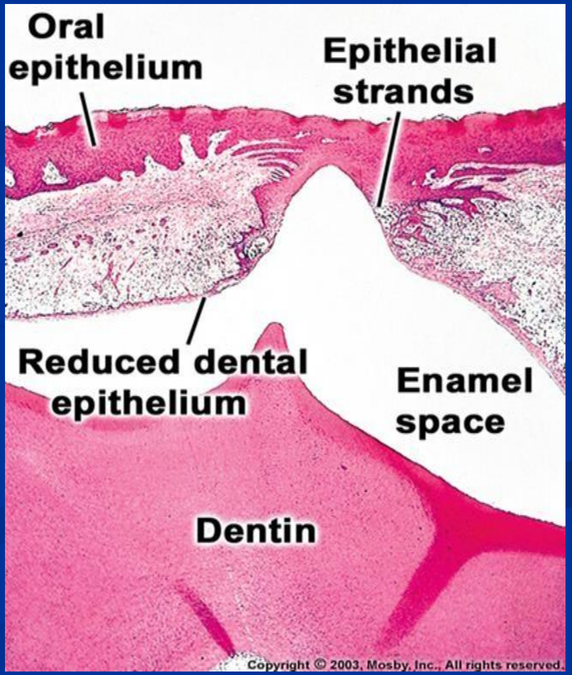
- The Gubernacular canal contains remnants of dental lamina and connective tissue termed as **Gubernacular cord.**

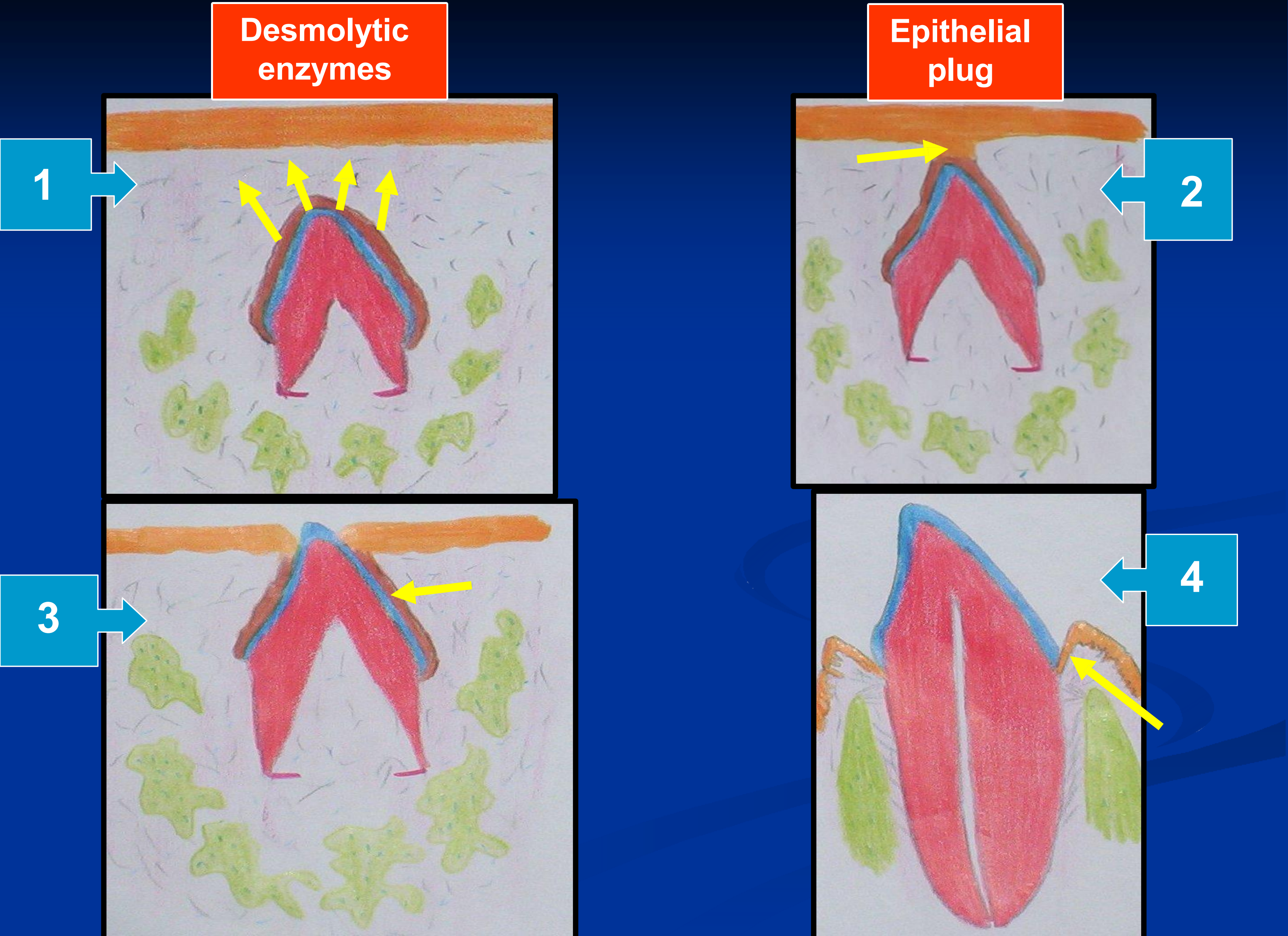
- Both Gubernacular canal and cord provide the directional path for eruption of the permanent tooth.

I- Changes in Tissues Overlying Erupting Tooth

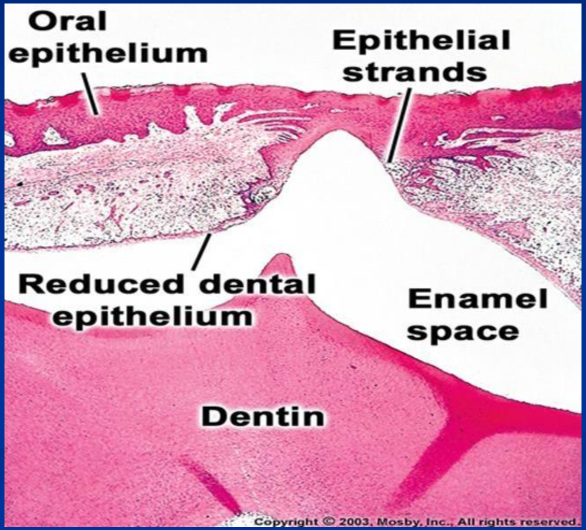
- Bone resorption in roof of crypt by osteoclasts
- Degeneration of C.T. between R. D. E. & oral epith.
- Epithelia proliferate & fuse forming a central mass.
- Central cells of the epith. mass degenerate
- Formation of an epithelial lined canal for the tooth to erupt




Desmolytic enzymes
Epithelial plug
1
2
3
4

- **Tooth eruption isn't accompanied with pain due to**
  1. degeneration of the overlying connective tissue
  2. decrease in the no. of bl.vs & break down of nerve fibers

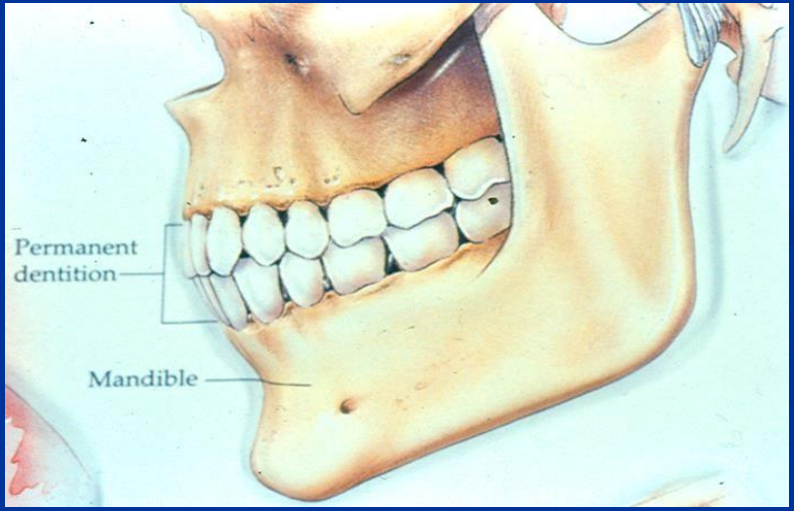


- The fusion of the reduced enamel epithelium with that of the oral epithelium cause stretching & thinning of the oral mucous membrane (**blenching of the mucosa** overlying the erupting tooth).

- An epithelial lined canal is formed by degeneration of the central cells , through which the tooth erupts **without bleeding**

3) Eruptive (functional) phase:

- Begins….when tooth reaches <u>occlusal plane</u>



- Ends……at the end of life span of the tooth

Pattern of tooth movement in the post-eruptive phase

- During this phase the tooth moves to :

*Maintain its position while the jaws increase in height………… ……………….by axial movement

*Compensate for occlusal wear……axial movement

*Compensate for proximal wear……mesial drift

Mechanism OR Causes of Eruption

Theories:

- Root formation theory
- Bone formation or Remodeling theory
- Dental follicle cell proliferation theory
- Periodontal ligament traction theory
- Pulpal Or Vascular pressure theory

- **The most appearing important causes are:**

  1 -Elongation of the root

  2 -Modification of alveolar bone

  3- Modification of periodontal ligament

These events are coupled with changes overlying the tooth that form the eruption pathway

…….remember how…….

References:

1- Essentials of Oral Histology and Embryology, James K Avery; 3rd edition, Mosby. Chapter 6, Pages81- 95

2- Tencate's Oral Histology. Development, structure and function, Antonio Nanci, 8th edition,2008, Mosby Inc. Chapter 10, Pages 268-289

Thank you